Homework 7: DCGAN

CS 1470/2470

Due Dec 6th, 2019 at 11:59pm

1 Conceptual Questions

1. What is the discriminator loss? What is the generator loss? What are they each measuring? (2-4 sentences)

   Answer: The discriminator loss and the generator loss can be written as:

   $$L_D = -\mathop{\mathbb{E}}_{x\sim p_x}[log(D(G(x))] - \mathop{\mathbb{E}}_{z\sim p_z(z)}[log(1 - D(G(z))]$$

   $$L_G = \mathop{\mathbb{E}}_{z\sim p_z(z)}[1 - log(D(G(z))]$$

   The generator loss L_D measures the probability of fake images identified as fake, and the discriminator loss measures the sum of the probability of fake images identified as fake and real images identified as real.

2. Why do we alternate between training the generator and training the discriminator? (2-4 sentences)

   Answer: We could use two tapes to record the backward propagation order for the generator and the discriminator. We could calculate the loss for generator first to renew the parameters in generator, and repeat the same strategy for the discriminator.

3. One common problem encountered during GAN training is mode collapse. Explain what mode collapse is and why it occurs. (2-4 sentences)

   Answer: Mode collapse means that the generator collapses to the output which it could be realistic enough to fool the discriminator. As a result, all of the output collapses to one or several same outputs. The reason can be attributed to the loss of generator asked to generator one output that look realistic enough, but it did not ask every output to be realistic enough.

4. Recall that the generator's objective is to minimize the following loss function:

   $$L_G = \mathop{\mathbb{E}}_{z\sim p_z(z)}[log(1 - D(G(z))]$$

The authors of the original GAN paper propose using a slightly modified loss function for training the generator. State the function and describe why it is favorable, in your own words. (2-4 sentences)

Answer: In their paper, instead of minimizing $L_G = \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]$, they chose to minimize $L_G = \mathbb{E}_{z\sim p_z(z)}[\log D(G(z))]$ to avoid gradient saturation. This loss function is more favorable since it provides much stronger gradients early in learning.

5. We use Frechet Inception Distance (FID) to evaluate the performance of our model. What does FID measure? What are some of its limitations? (2-4 sentences)

   Answer: Frechet inception distance measures the distance between the distributions of output features. In other words, it can measure the quality of output images compared with real images as an index for GAN performance measurement. The Frechet inception distance approximates the activation of real and generated images as Gauss distribution which could be too coarse to measure subtle details.

2 Ethical Implications

1. Question 1 Lawmakers are increasingly interested in regulating deepfakes; for instance, this is an analysis of the DEEPFAKES bill in the US house. (California and Texas have passed related bills.)

   (a) A lawmaker has reached out to you to explain to them how deepfakes "work" – they have read some news articles that talk about something called GANs, but want to know more. Explain in laypersons' terms what a "GAN" is, and how it can be used to produce something like a deepfake. (4-6 sentences)

   Answer: GAN is a subclass of generative model which means that it can learn from real images and produced new and fake images. The analogy of learning strategy can be thought as there are two players, one generator and one discriminator. The generator wants to draws something that is fake but real enough to fool the discriminator. The discriminator wants to be smart enough to distinguish fake images and real images. When the generator and the discriminator are well-trained enough, the generator could produce something that never exist but looks indistinguishable from real images, such as human face(deepfake).

   (b) Suppose you've just published a paper describing an algorithm where, if given a video, the algorithm can tell whether or not it is a deepfake. You want to use this new development to reduce potential harms of

deepfakes; in what settings might this algorithm be used, and by whom? What (potentially non-technical) considerations need to be made when using this algorithm (e.g. free speech), and how will those decisions be made? (3-4 sentences)

Answer: This tool could be used by some authority agencies to identify the deepfake. It has to be accurate enough, user-freindly enough, and well-included enough to accurately identify the multi-source fabricated data such as images, voices or videos with ease of use. However accurate this algorithm would be, there is a certain probability that it provides wrong decision, hence we should not make our judgement solely lean on this algorithm. We should take this decision for reference and combined with our experience to make the final decision.